Assessment tool CO2 - AT1

Name: Kaviya . R

Reg NO: 192521075

Course Code: CSA0612

Course Name: Design and Analysis of Algorithm.

Analytical Problem solving:

53)

Problem:

$A = [61, 24, 13, 47, 32, 89, 53]$

find the target element 89 using the linear search algorithm.

Interpret the problem:

Linear search checks each element one by one from the begining of the array until the target element is found on the array ends.

Dataset size (n) = 7

Target element = 89

sequential search process:

step	element compared	Result
1	61	Not found
2	24	Not found
3	13	Not found
4	47	Not found
5	32	Not found
6	89	found.

Number of comparisons:

comparisons Made:

$61 \rightarrow 24 \rightarrow 13 \rightarrow 47 \rightarrow 32 \rightarrow 89$

Total comparisons = 6

The target 89 is found at index 5 (6th position)

Time complexity Analysis:

Best case : O(1) - Target is the first element.

Average case : O(n) - Target is somewhere in the Middle.

Worst case : O(n) - Target is the last element or Not present.

Verification :

The search stops when 89 is found.

Result :

Target element = 89

position = 6 th element

Index = 5

Comparisons = 6

therefore, the linear search result is correct.

efficiency Interpreation :

since the target appears near the end of the dataset, linear search must compare almost every element finding.

*) More comparisons are required.

*) search takes longer than when the target is Near the beginings.